1.4 Energy systems: fatigue and recovery	**1.4.1**	**Knowledge and understanding of the concepts of energy, with specific reference to physical activity and sport.**
	1.4.2	**Understanding of the forms of energy, processes by which it is regenerated, how depletion occurs and the recovery process.**
	1.4.3	**Forms of energy to include: mechanical, electrical, potential, chemical and kinetic. The role of energy as adenosine triphosphate (ATP) in muscular contraction and the use of phosphocreatine (PC), glycogen and fat as sources for ATP re-synthesis.**
	1.4.4	**The characteristics and physiology of the three energy pathways (ATP-PC, glycolytic and aerobic).**
	1.4.5	**The characteristics of the three pathways with regards to ease and speed of ATP production, the force of contraction that each will support, the intensity and duration of exercise supported by each as the dominant energy provider, and the regeneration of ATP for each pathway.**
	1.4.6	**The principle of the energy continuum when based around athletic running events.**
	1.4.7	**Use of the continuum as a medium to support understanding of the joint and collaborative role of the three energy pathways in physical activity.**
	1.4.8	**Positioning of athletic running events on the energy continuum.**
	1.4.9	**The concept of fatigue and factors that contribute to fatigue: energy depletion, dehydration and the build-up of waste products (including an exploration of the role of lactic acid in performance).**
	1.4.10	**Stages of recovery and their application to specific physical and sporting contexts.**
	1.4.11	**The fast component of recovery and re-phosphorylisation; the speed and rate of phosphogen replenishment.**
	1.4.12	**The slow component of recovery; the oxidation of lactate (removal of lactate and H^+), replenishment of energy stores and the two-hour window of opportunity: rehydration, physical cooling and thermoregulation; the 48-hour window of opportunity: resaturation of myoglobin, re-synthesis of protein, glycogen and carbohydrate (CHO); exercise induced muscle damage (EIMD) and delayed onset muscular soreness (DOMS).**
	1.4.13	**EPOC (excess post-exercise oxygen consumption), and the stages of recovery.**
	1.4.14	**Understanding of how the energy systems respond acutely to the stress of warming up/priming exercise.**